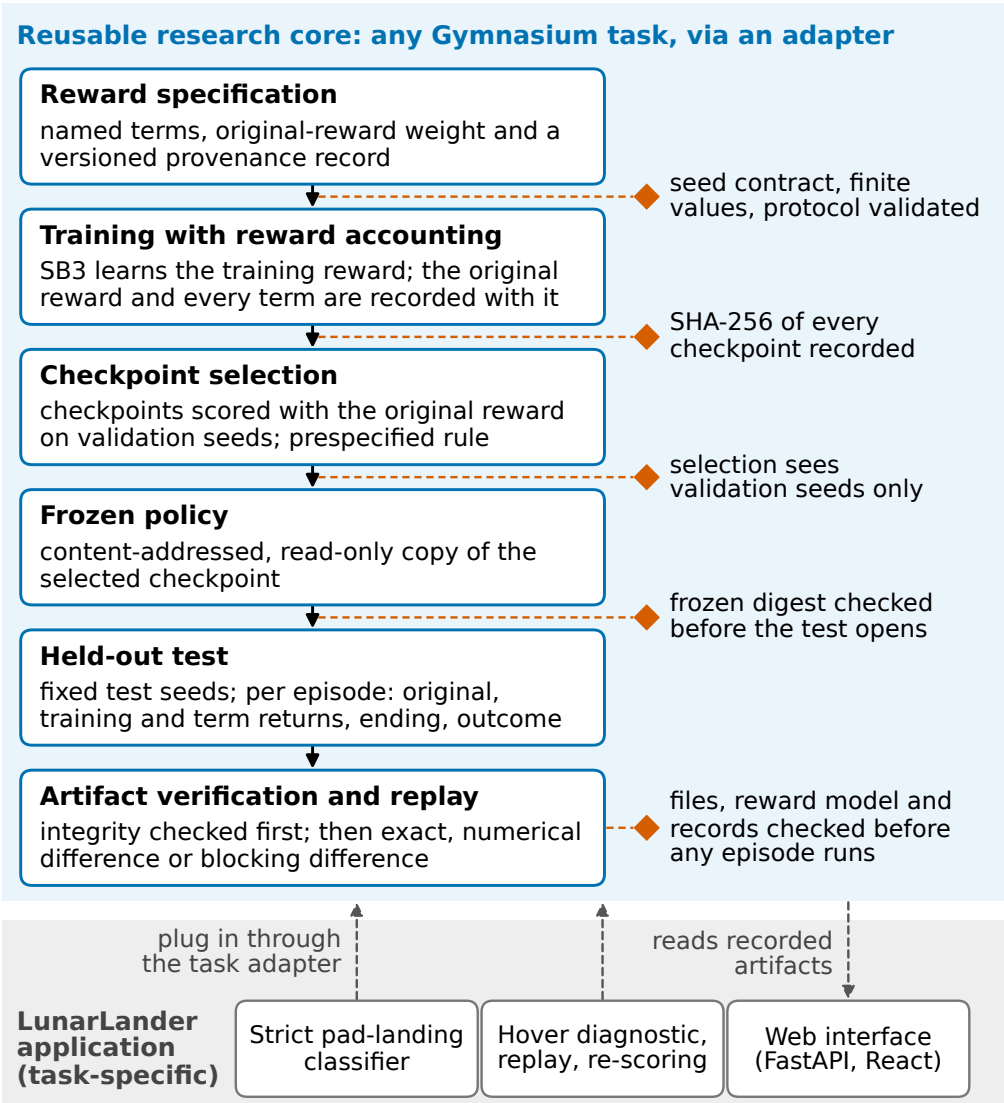
Reusable research core: any Gymnasium task, via an adapter
Reward specification
named terms, original-reward weight and a versioned provenance record
seed contract, finite values, protocol validated
Training with reward accounting
SB3 learns the training reward; the original reward and every term are recorded with it
SHA-256 of every checkpoint recorded
Checkpoint selection
checkpoints scored with the original reward on validation seeds; prespecified rule
selection sees validation seeds only
Frozen policy
content-addressed, read-only copy of the selected checkpoint
frozen digest checked before the test opens
Held-out test
fixed test seeds; per episode: original, training and term returns, ending, outcome
Artifact verification and replay
integrity checked first; then exact, numerical difference or blocking difference
files, reward model and records checked before any episode runs
plug in through the task adapter
reads recorded artifacts
LunarLander application (task-specific)
Strict pad-landing classifier
Hover diagnostic, replay, re-scoring
Web interface (FastAPI, React)